

Human-Centric Industry 5.0 Reference Architecture

Coupled Machine and Human Digital Twins for Sustainable Task Allocation

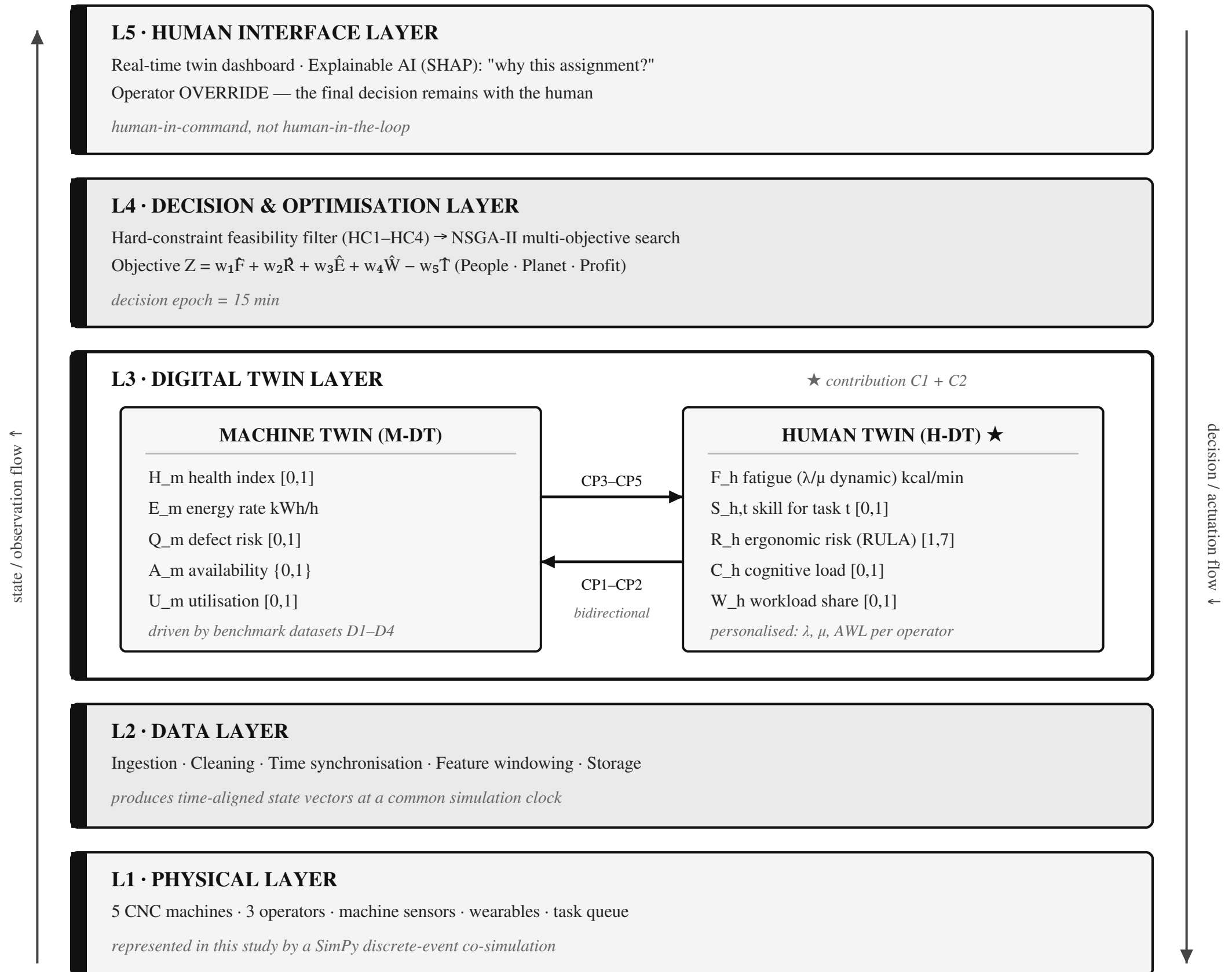


Fig. 1. Five-layer reference architecture. The novelty is L3: a machine twin and a quantified, personalised human twin exchanging state bidirectionally, both feeding a single constrained multi-objective decision layer.